

PIA 410 Course Outline

PIA 410 – GIS for Project Appraisal

Where the project is, who it reaches, and what the distance really is.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 409, whose exposed values this course can finally assess, and PIA 405 for the beneficiary counts that rest on a population layer.
Audience	Appraisal economists who have to state where a project is and who it reaches, and reviewers who have to judge whether a spatial claim is supported
Pass mark	60 %

What this course is

Three questions in an appraisal are spatial and are routinely answered without any spatial analysis at all. Where is the project? Who does it reach? And what is the distance actually being saved?

PIAS implements the machinery: 6 modules, 2262 lines and 73 functions across geometry, overlays, accessibility and spatial need, with 12 configured layers and a 61-assertion test file. The geometry kernel is hand-written – neither ArchGDAL nor LibGEOS is enabled – which is worth knowing before relying on a topology operation.

The register tells a different story. 8 of the 11 projects are recorded as a single point: no length, no area, no footprint. A point cannot be overlaid on a flood polygon, cannot have a service area, and cannot answer any of the three questions. And `Overlays.hazard_exposures` is implemented while every exposed value in PIA 409's hazard screening is an asset value rather than a spatial overlay.

What a graduate can do

State what a coordinate reference system is and why EPSG:4326 is not a projection; compute a great-circle distance and say why it does not need one; recompute a stored length from its geometry and detect a coarse alignment; say what a point geometry can and cannot support; perform a population overlay and state its dependence on the population layer's vintage; compute an accessibility indicator

at the Rural Access Index threshold and report it against the national target; distinguish a straight-line distance from a network distance and quantify the difference; compute a travel time saving from a speed table and say what the speeds assert; identify a spatial need gap; and state what a spatial result rests on.

What this course does not claim

This course does not teach a GIS package. There is no QGIS, no ArcGIS and no PostGIS here. Everything is done from coordinates with the standard library, because the point is to understand what the operations compute rather than which menu they sit under.

It does not supply spatial data. The 12 configured layers are layer definitions, and the register's geometries record their source as "SAMPLE_DATA". Where a real population, road or hazard layer comes from is a data question, and 08_Data/External_Data_Sources.md names the open datasets that would serve.

It does not do network routing. A travel time here comes from a distance and a speed table. Genuine routing needs a connected network with turn restrictions and is a different subject.

It does not project. The system works in geographic coordinates on a sphere, which is adequate for distance and area at this latitude and is not adequate for everything. The course says which is which.

Learning outcomes

1. State what a coordinate reference system is, and what EPSG:4326 is not.
2. Compute a great-circle distance and say why no projection is needed.
3. Recompute a stored length or area from its geometry.
4. Say what a point geometry can and cannot support.
5. Assess whether a geometry supports the claims made of it.
6. Perform a population overlay and record its layer and vintage.
7. Compute an accessibility indicator at a stated threshold.
8. Report an accessibility result against the Rural Access Index.
9. Distinguish a straight-line distance from a network distance.
10. Compute a travel time saving from a speed table.
11. Produce a hazard overlay and a spatially derived exposed value.
12. State what every spatial figure rests on.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	Where the project is	1	1.25	–	1	Quiz M01
M02	Coordinates, and what they are not	1	1.50	–	2	Quiz M02
M03	Distance on a sphere	1	3.25	Lab 01	3	Quiz M03
M04	What a point cannot do	1	1.50	–	4	Quiz M04

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M05	Reconciling a geometry	1	3.50	Lab 02	5	Quiz M05
M06	The population overlay	1	1.50	–	6	Quiz M06
M07	Accessibility, and the Rural Access Index	1	3.50	Lab 03	7	Quiz M07
M08	Straight line against network	1	1.50	–	8	Quiz M08
M09	Travel time and the speed table	1	3.50	Lab 04	9	Quiz M09
M10	Spatial need, and where the gaps are	1	1.50	–	10	Quiz M10
M11	The hazard overlay PIA 409 needed	1	3.50	Lab 05	11	Quiz M11
M12	Writing the spatial section	1	1.50	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Recompute every geometry, and say what each supports	2.0	A verified geometry table
2	List the spatial claims, and test each against the geometry	2.0	A claims assessment
3	Compute an accessibility gain at the Rural Access Index threshold	2.0	An accessibility result, labelled as RAI
4	Compute a travel time saving, and test the assumption that carries it	2.0	A travel time saving with its sensitivity

Lab	Title	Hours	Deliverable
5	Run the overlay PIA 409 needed, and report honestly where you cannot	2.0	A spatially derived exposure

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	Every geometry, checked	25	Marked on the recomputation and on the NOT APPLICABLE distinction
A2	The claims that outrun the geometry	25	Marked on distinguishing false from unsupported
A3	Accessibility and time	25	Marked on the passability judgement and on the elasticity
A4	The overlay PIA 409 needed	25	Marked on the derivation and on reporting the points honestly

Final project

A complete spatial section for one project: the geometry with its type, its coordinate reference system and its vertex count; the length or area recomputed from the coordinates and reconciled against the stored value; an assessment of whether the geometry can support the claims the appraisal makes of it; a population overlay with the layer and its vintage recorded; an accessibility result at the Rural Access Index threshold, reported against the national indicator; a travel time saving computed from the speed table with the speeds stated as assumptions; a hazard overlay producing an exposed value to replace the asset value PIA 409 used; and a closing paragraph stating what every spatial figure rests on.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.